



RESEARCH ARTICLE

REVISED **A digital twin enabled wearable device for customized healthcare**

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Abstract

Background

The traditional healthcare process centers on the hospital rather than the individual patient. The demand for continuous monitoring is increasing with the increasing proportion of patients with chronic diseases and the elderly. Wearable medical devices have brought medical monitoring into the Internet age. To improve the devices' adaptability, this research proposes a combination between digital twin (DT) and wearable medical devices is proposed to provide personalized wearable medical devices and personalized healthcare efficiently.

Methods

A DT-enabled smart system is proposed for personalization in the design, manufacturing, and data tracking of a healthcare device prototype. DTES is a proposed DT-based personalized service system framework. Through the digital mapping of user physical data, manufacturers and third-party service providers can quickly respond to user needs and complete the design process quickly in the digital space, thereby supporting manufacturing in the physical space. A case study is made for three healthcare monitoring scenarios: rehabilitation training, wheelchair, and human fall. Based on computer-aided design and additive print, a triaxial vibration collection bracelet with a simple Internet of things mode is designed and manufactured in personalization.

Open Peer Review

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Results

The bracelet shows great application ability in this case study, including design, manufacturing, and remote connection. 10 groups of data were recorded in each scenario. In rehabilitation training and wheelchair experiments, the average values of correlation coefficient between models and the actual data are 0.991 and 0.749 respectively. In human fall experiment, the motion signal parameters of the user and movement pattern were clearly identified. These results provide the basis for applications in different scenarios.

Conclusions

The device is representative, with good personalization and health monitoring performance, and has excellent potential for large-scale application. DT will provide a new feasible solution for the realization of personalized medicine.

Keywords

Digital Twin, Personalization, Internet of things, Smart manufacturing, Personalized healthcare

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REVISED Amendments from Version 2

1. In Introduction part, we have elaborated on the significance of introducing digital twin into the design and manufacturing framework of wearable medical devices in the article, as well as their comparison with traditional wearable devices.
2. In the Case study part, we provided more detailed experimental parameters, including instrument parameters as well as material parameters and preparation process for 3D printing. This helps to reproduce and validate the experiment.
3. In the Conclusion part, we have conducted a detailed evaluation of the effectiveness of DTES in a dynamic personalized scenario, highlighting some shortcomings of the framework and future solutions. At the same time, it also expands the limitations of the research, as well as the long-term research potential and goals.
4. Additional citations of relevant literature have been added.

Any further responses from the reviewers can be found at the end of the article

Introduction

Recently there has been a dramatic growth in number of patients with chronic diseases due to population aging (Ferrucci *et al.*, 2020). The time and economic cost of healthcare are high for patients with limited access to health services (Guk *et al.*, 2019). To solve this problem, it may be beneficial to transmute healthcare monitoring from hospital-centric to patient-centric. That would mean a demand in real time personalized healthcare management. As wearable medical devices began to emerge, remote monitoring and diagnostics gradually permeated various areas of healthcare (Erol *et al.*, 2020).

Diverse disease types and physiological structures of patients fuel the application of a variety of wearable medical devices (Lou *et al.*, 2020). A natural issue that arises with wearable devices is the needs based on individual patients, a solution to this could be customized wearable devices, as a method to feed specific healthcare demand. Now consider extending this idea to mass application: a personalization service model covering the whole product cycle. It will allow mass personalization and traceability for wearable medical device. Digital twin is a pioneer for personalization, supported by Internet of things, Big data, and Artificial intelligence (Aheleroff *et al.*, 2020a). The synchronization of physical and digital healthcare devices will have substantial improvements on design, manufacturing, and healthcare services (Schleich *et al.*, 2017). Through the collection of important physiological data, digital twin (DT) helps provide personalized wearable devices and accurate customized care for patients (Tang *et al.*, 2024). Unlike traditional wearable devices, DT's virtual twin model not only maps user physiological data in real-time but also dynamically optimizes device design parameters through machine learning. For example, long-term user usage data can be fed back into the digital space, triggering design iterations (such as adjusting the structure of rehabilitation equipment based on muscle activity patterns) and thereby achieving adaptive

personalization. This closed-loop feedback mechanism is a significant breakthrough in existing technology.

This study makes an original contribution combining DT services and wearable medical devices. Through digital mapping of device input and output, this research provides tailored health support device in high efficiency. It should make a significant contribution to personalized healthcare development. After the introduction, the following section represents recent related literature.

Literature review

According to the summary of Tasnim *et al.* (Tasnim *et al.*, 2018), the fundamental functions of wearable medical device are: 1) comfort to the human body; 2) seamlessly extract and interpret biological information. The first function is fulfilled by design and manufacturing structure, and the second one by Internet of things technology. To achieve a personalized process in wearable medical devices, this section reviews the related work on DT for personalization, and the interaction between wearable devices and personalized healthcare is also considered.

DT for personalization

In the past decades, digital manufacturing has created tremendous value for industrial production. The rapid development of information and communication technology and computer-aided technology has brought significant improvements in all fields of industry. The latest development of these technologies focuses on the DT (Liu *et al.*, 2021). The DT is usually described as the overall structure of physical entities and their virtual counterparts and the data connection between them (Jones *et al.*, 2020). It is the digital form of physical objects, enabling data to be seamlessly transmitted in the physical and virtual worlds so that computer technology can be used to explore, manage entities, and improve their performance of entities (El Saddik, 2018).

In the personalized design and manufacturing system, the DT builds a digital model and virtual definition for the manufacturing system to help realize the digitization and automation in the design and manufacturing process. In the context of industry 4.0, the DT provides a good foundation for mass personalization (Aheleroff *et al.*, 2020a). At the same time, physical objects such as materials, personnel, and equipment can also be built into digital form to get unified and efficient management. Furthermore, this allows for the realization of additional intelligent maintenance, real-time remote monitoring, state prediction, and other advanced functions (Aheleroff *et al.*, 2021).

Many cases show that DTs can handle many personalized products and services cost-effectively, thus providing unique value for personalization. For example, simulation space is provided in personalized product design to help engineers and customers intuitively develop products (Mourtzis *et al.*, 2022). It can be used to build physical information systems to help personalized production (Park *et al.*, 2020), and it

can also be used in applications to build unique personalized monitoring and remote-control systems (Aheleroff *et al.*, 2020b).

Personalized healthcare

Continuous and effective monitoring of chronic diseases will substantially improve patients' quality of life and reduce risk. With the progress of technology and the development of Internet of things technology, medical and health care services have gradually shifted from hospital-centered nursing to individual-centered personalized services (Yan *et al.*, 2013). Personalized medicine can provide personalized medical care services for individuals to provide long-term monitoring and effective treatment for patients and improve their lives (Nardini *et al.*, 2021).

Through personalized medical treatment, various clinical procedures can be replaced by remote real-time monitoring, such as heart rate monitoring, pressure monitoring, blood glucose detection, and so on (Jagadeeswari *et al.*, 2018). This means that medical monitoring scenes can be expanded from hospitals to more diverse scenes, such as homes. At present, there are many good application cases of personalized medicine. For example, by integrating devices and services, the intelligent medicine kit (Yang *et al.*, 2014) can provide regular and quantitative personalized drug dispensing integrated with home medical services. A wireless body area network (WBAN) can measure the temperature distribution, heartbeat, and physical falls of users through the configuration of wearable sensors to realize automatic remote monitoring (Wu *et al.*, 2017).

In addition to personal applications, personalized medicine can extract physiological data of multiple people at the same time, promote the application of extensive data analysis in the health care industry, and make up for the lack of medical system information (Wang *et al.*, 2018). Of course, this process requires strict formats, protocols, and collaboration networks (Hulsen, 2021). At present, personalized medicine has not been fully realized. The standard of personalized medical devices is still challenging to establish and popularize (Guk *et al.*, 2019). At the same time, in the face of industrialization, the cost is still a significant problem (Tasnim *et al.*, 2018). Still, it is undeniable that personalized medicine is a promising direction in health care.

Internet of things enabled wearable devices for healthcare

The development of mobile devices, the ubiquitous Internet, and emerging network technologies (such as 5G) have provided vital support technologies for the extensive deployment of the Internet of things (Shafique *et al.*, 2020), enabling connectivity between physical objects and the Internet. It can be said that so far, the application of the Internet of things has been seen everywhere. In recent years, wearable devices have formed a particular market segment due to the combination of sensing and computing capabilities, especially having a far-reaching impact on the healthcare industry (Hiremath *et al.*, 2014). Wearable devices constantly collect and process physiological or environmental data related to human health and then carry out data sensing and

driving. Thus, users' physiological and health status will be effectively monitored (Sheth *et al.*, 2018). Through artificial intelligence and cloud computing, wearable devices can use data seamlessly to diagnose diseases and build intelligent medical systems (Mansour *et al.*, 2021).

In the field of health care, wearable devices based on the Internet of things have brought many substantive applications, such as sports trackers (To & Mahfouz, 2013), arrhythmia monitoring (Yamaç *et al.*, 2022), epilepsy treatment (Teurlings *et al.*, 2022), sleep health care application (Yang *et al.*, 2021), intelligent health monitoring platform (Kang *et al.*, 2018), and so on. These applications are diverse, involving all aspects of the health care field. Predictably, the application of the Internet of things in medical wearable devices will gradually increase and become the fundamental device of mainstream medical care in the future (Indrakumari *et al.*, 2020). However, some medical information privacy and security issues still need to be solved (Somasundaram & Thirugnanam, 2021).

A DT-enabled smart device system

Wearable medical device manufacturing is a challenging and highly regulated field (Lepasepp & Hurst, 2021). Due to direct interaction with patients, medical device developments are highly dependent on end users, that is, aged, people with disabilities and/or special needs, and their caregivers (Shah *et al.*, 2009). Given this characteristic, one is left to reflect on how the wearable medical device manufacturing is supported by the technological enhancements brought about through DT. DT supports lean manufacturing and helps improve safety and reduce production time and cost through transparency optimization (Bazaz *et al.*, 2019). Some of the more functions introduced by DT are data centralization and traceability, making DT a promising technology for wearable medical device.

At present, the application of digital twins in wearable devices mainly focuses on the assembly of sensors and communication processing (Yu *et al.*, 2022), and there is still a lack of guidance for wearable devices at the design and manufacturing levels. This study proposes a DT-enabled smart device system for involving various types of patients into personalization and healthcare service of their wearable devices. It is an acceptable and theoretical structure. Through efficiency management of information of all participants in the manufacturing, the system produces highly personalized devices conveniently. Based on the Internet of things in wearable devices, the function of continuous tracking is also provided.

System structure

The DT-enabled system (DTES) is a personalized service system covering the whole product cycle. It connects all participants in the design and manufacturing system through cloud services. In DTES, user data collection, design, manufacturing, third-party services, and other parts have data mapping and complementary cloud services. The manufacturer will centrally manage data to ensure that the personalized production process can be carried out smoothly among all participants. The structure of DTES is shown in Figure 1.

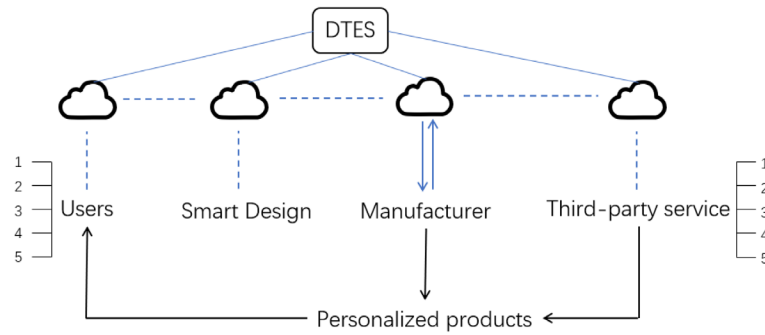


Figure 1. Digital twin enabled system (DTES) structure.

Each user's DT consists of two kinds of personalized data: the personalized parameters for design, and the data generated by the long-term use of the product. User parameter data for design needs to be input, by users. Such data will be extracted from the cloud and generate relevant computer aided design (CAD) design files, becoming the input files of personalized manufacturing. This process is the basis of personalized design and manufacturing. The data generated during long-term use include physiological data, state parameters, *etc.* These data need to be monitored for a long time and further applied. Therefore, in DTES, we will store and archive these data in the cloud, which the application system will retrieve at any time. For some additional functions and components in personalized products, it is difficult for manufacturers to re-develop and produce them because this usually means high costs. Cooperation with third-party service providers allows manufacturers to build the personalized products they need quickly. In this process, DTES can act as a "bridge" between the two through cloud services. Manufacturers can quickly integrate the information of third-party services and combine relevant services to shorten the time of personalized manufacturing.

After all participants in the manufacturing system are connected through cloud services, manufacturers can uniformly manage information and efficiently produce personalized products. For manufacturers, different service information is transparent, so product design and production strategies can be adjusted in time. The content of the service can be interfered with at any time, which makes the system flexible. Therefore, the system can have great potential to produce personalized products with high degrees of personalization efficiently.

Digital twin enabled device

The capacity of data collection, transmission, and storage is highly demanded for wearable device due to medical functions of monitoring and/or support (Degerli & Ozkan Yildirim, 2021). The convenience and connectivity are essential. Therefore, application of Internet of things technology in wearable devices for healthcare is proposed in DTES.

It is needed to maintain a continuous and stable remote connection to make sure healthcare can be carried out smoothly.

DTES endows sensors with signal transmission elements in the design and manufacturing process and uploads physiological data in time through Internet or short-distance communication network to realize real-time monitoring and remote-control functions.

Unlike IoT devices under traditional manufacturing systems, wearable devices based on DTES not only utilize IoT to enhance information exchange in terms of functionality but also apply digital twin mapping to manufacture devices flexibly. Specifically, users' physical information is mapped into the digital space, such as size parameters, wearing parts, requirements, *etc.* The design products in the digital space will be manufactured in the physical space through 3D printing and assembly of third-party components. In this way, DTES will achieve more flexible and personalized wearable device manufacturing based on traditional IoT devices.

Traceability and trackability

Users' physiological data needs to be collected, transmitted, and effectively used. Figure 2 illustrate a healthcare data traceability and trackability model consists of five progressive hierarchies: the physical layer, sensor layer, communication layer, network layer, and application layer. They are all indispensable due to a bottom-up hierarchy design.

The physical layer defines the means of users and raw data produced by their physiological activities. These original signals are physical signals or chemical signals (e.g. electrocardiogram (ECG) signals, blood glucose).

The sensor layer is a variety of physiological signal acquisition devices responsible for converting the user's original signal into a digital signal. The sensor layer refers to not only the sensor itself but also the functions of filtering, amplification, signal receiving, and sending. Data can be stored and applied only by converting physiological signals into digital signals.

The communication layer refers to the signal transmission link, which includes a gateway, wireless network, and other infrastructure. The function of the communication layer is to transmit signals collected by the equipment to

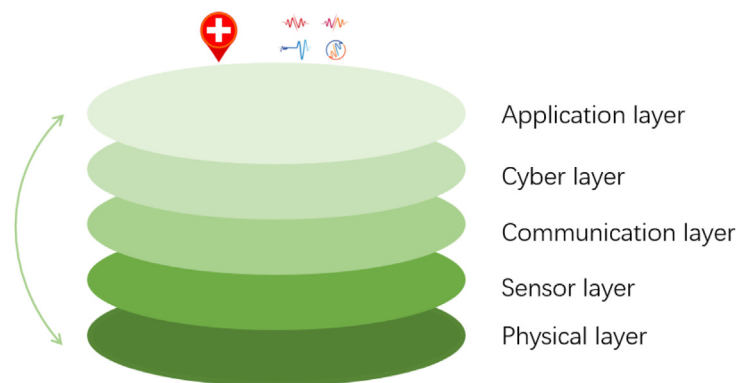


Figure 2. A healthcare data traceability and trackability model.

the outside world through the network to realize the flow of information.

The cyber layer includes cloud storage and processing. Compared with solid-state storage, data storage in the cloud is more convenient, efficient, and large capacity. At the same time, the user's personalized database can be established for calling at any time to get personalized applications.

The application layer is the destination of data transmission. It includes hospitals, emergency services, and various health analysis software. Users' personalized data is retrieved from the cloud and applied into physiological analysis, early warning, generating physical examination reports, *etc.*

This references model establishes an effective data transmission channel between the physical layer that generates physiological data and the application layer that applies physiological data. As a vital function of user-side healthcare services, traceability and trackability enables physiological data to be applied in real-time, maximizing the value of signals. At the same time, devices based on the Internet of things can ensure that the physiological data of multiple users are uploaded and analyzed simultaneously. This provides a good background for the realization of personalized medicine.

Case study: personalized healthcare

The care of the elderly and the disabled poses a challenge globally (Feng, 2019). The elderly and the disabled bear a higher risk from activities in their daily life. The enormous pressure that long-term care brings to family members and the government needs to be reduced. Therefore, this study decides to develop personalized smart device to assist in long-term care of the aged and disabled.

Case description

The needs of medical care are diverse. People need all kinds of medical care since they have different physiological

conditions. Especially for older or disabled people, their risk of being damaged by accident is more obvious, so more medical care sources should be provided for them. However, labor resources are facing a shortage in long-term care (Wyonch, 2021). Moreover, modern care methods which improve health data accessibility and delivery are critical. A reasonable solution is to use personalized wireless equipment to monitor physiological data for early warning and replace manual inspection. As shown in Figure 3, for covering most situations in the daily care of older people and disabled people, we choose three common daily scenarios with different motion modes to evaluate the actual performance of DTES in personalized medical care. More scenarios can be explored based on combinations of different modes to make usage scenarios more flexible.

The first scenario discussed in this study is human falls. The problem of falls in the elderly should not be ignored. Generally, the elderly have poor balance. At the same time, additional drugs, joint disease, and weak physical conditions will exacerbate this. As a result, they have a higher risk of falls. Research in 2020 (Ang *et al.*, 2020) shows that one-third of people over 65 falls yearly. For people aged over 60, fall deaths account for 40% of accidental deaths (Australian & New Zealand Falls Prevention Society, 2020). Furthermore, this ratio is higher for people older than 80, surpassing 50% (O'Loughlin *et al.*, 1993). Preventing falls is a social vision, but it can hardly come true. One possible solution is to detect falls in time so that fast help can be provided. It can decrease the severity of falls in people in long-term care, especially in bead house or home scenarios.

The second scenario discussed is a wheelchair. According to research in 2011 (Chen *et al.*, 2011), among 95 wheelchair users, 54% of them reported more than one accident in three years. Researchers develop methods to decrease accidents caused by wheelchairs. A danger detection system for wheelchair safety (Kim & Hasegawa, 2012) can detect accidents in wheelchairs and report warnings. The system can evaluate the type and severity of accidents to improve and

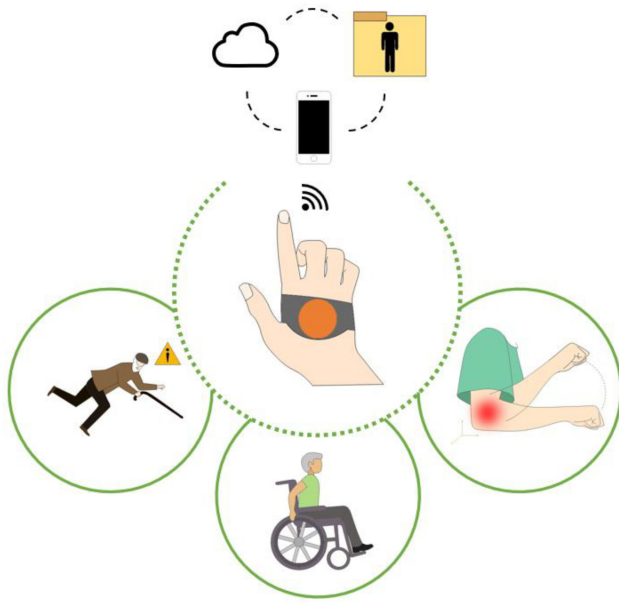


Figure 3. Schematic diagram of healthcare in different scenarios.

evaluate the safety of electric wheelchairs. Moreover, more adjustments related to vibration can be made to improve the comfort and safety of the wheelchair.

The third scenario discussed is rehabilitation training. According to neurorehabilitation medicine, rehabilitation training can recover damaged limbs differently. In this process, doctors need to evaluate patients' specific conditions so that rehabilitation plans can be made and adjusted. Sensors can collect data in rehabilitation training and improve evaluation and treatment (Kim *et al.*, 2022). Wearable sensors play an essential role in rehabilitation training by closely monitoring physiological data (Jiang, 2020). These data can help doctors build effective rehabilitation plans and evaluate sports ability and recovery conditions to improve training efficiency and achieve better rehabilitation effects. Cheaper and easier methods should also be developed for better implementing sensors and internet of things technologies in rehabilitation training.

Mobile devices, the cloud, and personal documents are connected, creating synergies that provide more possibilities for services. Cloud can provide cloud storage space to save more historical data. Moreover, through cloud transmission, data can be used by hospitals or third-party data processing companies for further processing. After data transmission, physiological vibration data will be saved on mobile devices such as cellphone or laptop and generates a document automatically. Personal documents will also be generated to provide personalized monitoring services and generate unique feedback. Of course, this process depends on efficient information transmission and data sharing protocols. Users' data privacy also needs to be protected.

Resolution

For all different scenarios, this study provides a continuous monitoring service. The bracelet and vibration sensor combination is a media between physiological vibration data and cloud or data analysis tools. It works in different scenarios when worn by users and collects vibration data through manual and automatic triggers (when a certain threshold is reached). Vibration data will be transferred to mobile devices in real-time through the sensor or transmitted directly to the cloud server through the gateway. The data transmission mode is Bluetooth, which is wireless. As a result, the data transmission process is convenient, steady, and quick. Scenario monitoring is not limited by physical condition as long as the user wears the device continuously. The design and manufacture of the bracelet need to be based on the size of the user. If the size does not match, the wearing experience and data collection effect will be affected at the same time.

System process

After clearing basic scenarios and resolution in the case study, we need to achieve the design and manufacture of the needed product, which is the intelligent triaxial vibration collection bracelet. The logic of the design and manufacturing process is shown in Figure 4. At the beginning of design and manufacture, users' size or other information is collected and input as characteristic parameters, which is the basis for the initial construction of products. Moreover, if additional components are needed, third-party service could be provided in future applications. This can increase the diversity of functions and designs to meet customers' more diverse personalized needs.

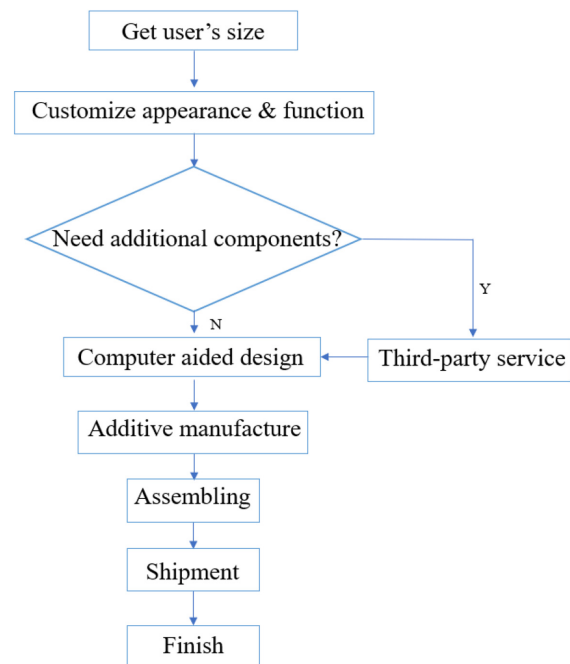


Figure 4. Design and manufacturing logic of digital twin enabled system.

CAD uses computers to help designers build, design, and optimize products to improve their productivity of designers (Sarcar *et al.*, 2008). CAD systems are widely used in processing and manufacturing, especially in the face of complex scenes, such as the design and manufacturing of innovative tools (Mikolajczyk *et al.*, 2018). Then based on CAD files, we used the additive manufacturing method to generate needed products. Additive manufacturing, also known as 3D printing, is well suited for applications and shape-based flexible innovation (Fuwen *et al.*, 2018). CAD files are the information source for generating incremental products (Mikolajczyk *et al.*, 2019). This feature can significantly improve the efficiency of personalized production, simplify personalized manufacturing, and control costs simultaneously. It is undoubtedly one of the leading production methods for the realization of mass personalization. In the future, after manufacture, we intend that components will be assembled with the main structure, generating the final products. Then personalized products are packaged and shipped to customers.

Personalized rapid design

The CAD document is essential for additive printing of products. Therefore, it is needed to construct the bracelet based on the needed size and shape from scratch and use a computer aided design to generate three-dimensional patterns for 3D printing. In this research, Solidworks software is used to generate 3D printed files. The design process of the 3D print document of the bracelet is shown in Figure 5.

In the beginning, the size of the user needs to be collected and input as parameters of design documents. As shown in Figure 5(a), The length and circumference of the user's palm are measured. The basic shape of the bracelet is a groove so that length and circumference can be converted into the length and height of the groove. The size of the bracelet will be designed based on these two parameters. This study chooses the EPH-V11 triaxial vibration sensor from Erbesd Instruments company as the sensor in a wearable device, with the following technical parameters: range $\pm 50g$ (XYZ axis), frequency response of 0.5-5000Hz, resolution of 0.0001g, linearity $\pm 1\%$ FS, sampling rate of 1000Hz. The sensor transmits data to the mobile terminal via Bluetooth 5.2 (compatible with iOS/Android), with a built-in storage capacity of 8GB (continuous recording time ≥ 72 hours). Signal

preprocessing includes real-time low-pass filtering (cut-off frequency 100Hz) and temperature compensation (-20°C to 60°C) to ensure data accuracy. The sensor is cylindrical, and its size is diameter *height = $48*34$ mm. There is a small hole with a diameter of 5 mm and a depth of 10 mm at the bottom to fix the sensor on different surfaces. As a result, on one of the long sides of the bracelet, we designed a circular base to place the vibration sensor. After inputting parameters, the base pattern will be stretched vertically by 2cm as foundation thickness. The size of the circular base is based on the diameter of the sensor, as shown in 5(b). In order to reduce the tangential stress of the vibration sensor so that it will not fall, part II is designed to fix the vibration sensor. When parts I and II are combined, they will become the appropriate size bracelet to fix the sensor on the palm, the final 3D design pattern shown in Figure 5(c). Then, part I (Figure 5c) of the bracelet was printed by 3D printer, and part II was printed based on part I. The wristband is made of thermoplastic polyurethane (TPU, Shore hardness 95A, tensile strength 45MPa) and printed using a Bambu Lab X1C printer. The key parameter settings are as follows: layer thickness of 0.2mm, nozzle temperature of 240°C , hot bed temperature of 80°C , and printing speed of 60mm/s. The high elasticity and wear resistance of TPU can adapt to different wrist sizes of users, while ensuring long-term comfort. Post processing includes ethanol acoustic cleaning to remove residual support materials. The process of intelligent bracelet design is efficient and low-cost. The personalized structure provides a good foundation for collecting vibration signals.

Internet of things of smart bracelet

Wearable devices based on internet of things enable people to understand their health status better and allow doctors to monitor patients' health and safety status remotely. The schematic diagram of the internet of things mode of the bracelet is shown in Figure 6. The wireless connection way can be Bluetooth or Gateway. Bluetooth enables the bracelet to connect to mobile equipment like a cellphone or tablet. This connection method is very convenient, and means that users can use their mobile devices to operate the collected data at any time, such as observation, storage, transmission, and so on, without restrictions of time and space. This sensor has its unique app (Wiser Vibe) for cellphones, which has some essential functions like saving, warning, or automatic

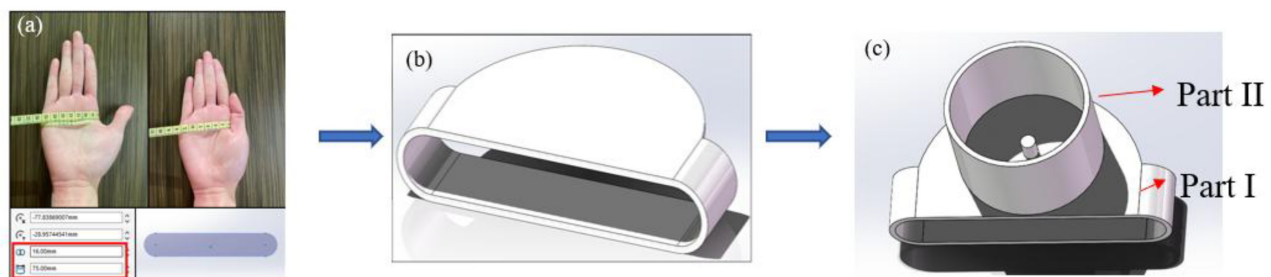


Figure 5. Design process of bracelet (a) collection of user's size (b) computer aided design (c) final 3D print document.

uploading. So we do not need to develop additional software on the mobile terminal.

Moreover, using gateway is suitable for connecting with computer. The vibration sensor is continuously connected with the gateway, and the data is transmitted to the computer any-time. Therefore, continuous measurement and early warning can be achieved. So this connection was used for long-term monitoring. Further analysis and applications also can be made by using all kinds of analysis tools to analyze the waveform of the vibration signal. At the same time, the data between mobile phones and computers can be interconnected at any time. Different transmission modes and analysis tools can be switched at will. Therefore, whether for uncertain or fixed scenarios, vibration data can be transmitted stably, which means that the internet of things of the bracelet has good elasticity.

Experiments and results

In this section, the smart bracelet is used to evaluate its performance of remote health monitoring in different scenarios through experiments. The experiment will be conducted separately based on different scenes: rehabilitation training, wheelchair and Human fall experiments. This study analyzes the data collected by prototypes in different scenes to judge the accuracy of data collection. The physical analyses are based on different experiments to obtain the mechanical basis in data analysis and study the scene behavior mode.

Rehabilitation training. The rehabilitation training [Zhu, 2022b] scenario is designed for those with elbow injury or disability. In this experiment, the subject was an adult male with normal elbow function. His forearm length (the distance from the palm to the elbow joint) is 35cm. The subject tested different moving directions of the elbow when wearing

the band and vibration sensor. The movements in three different directions are shown in Figure 7 and video S1 [Zhu, 2022c]. The positive direction of the forearm, 90 degrees to the inner side of the forearm, and vertical direction corresponding to the vibration sensor's X-axis, Z-axis, and Y-axis, respectively.

This experiment first constructs a digital sinusoidal model with dynamic parameters for further study. Then vibration signals after low pass filtered were compared with the model, and correlation coefficients were obtained. The relationships between signal and model can be judged objectively through correlation coefficients. For continuous vibration signals, the average value of the signal is 0. The correlation coefficient formula is equivalent to:

$$\rho_{xy} = \frac{\int_{-\infty}^{\infty} x(t)y(t) dt}{\sqrt{\int_{-\infty}^{\infty} x^2(t) dt} \sqrt{\int_{-\infty}^{\infty} y^2(t) dt}}$$

ρ_{xy} is the correlation coefficient of signal x and signal y.

In order to intuitively show the relevance between the vibration signal and the optimal model after waveform fitting, Figure 8 plots them in one chart. The blue line is the line of vibration signal, and the yellow one is the optimal model, which is the parameters group with biggest correlation coefficient value. The curves of the vibration signal and model have high similarity in the time domain. It can be preliminarily judged that the type of vibration signal curve is a sinusoidal signal curve.

Waveform fitting results are shown in Table 1. Amplitude A, velocity W, and correlation coefficient P of the dynamic model in 10 groups are saved as below. The range of

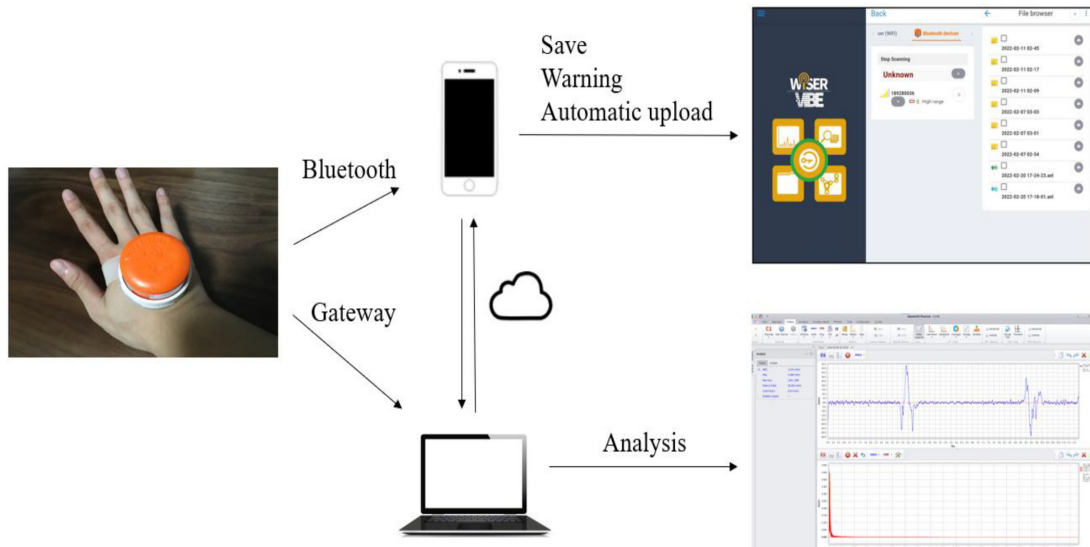


Figure 6. Internet of things of triaxial vibration collection bracelet.

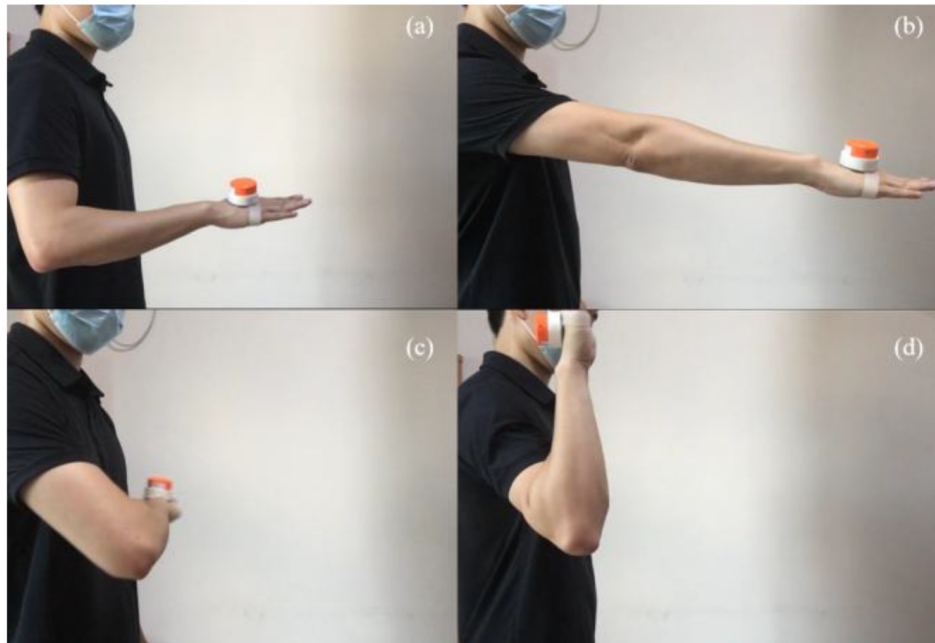


Figure 7. Joint movement in rehabilitation training experiment (a) initial posture (b) move along the X axis (c) move along the Z axis (d) move along the Y axis.

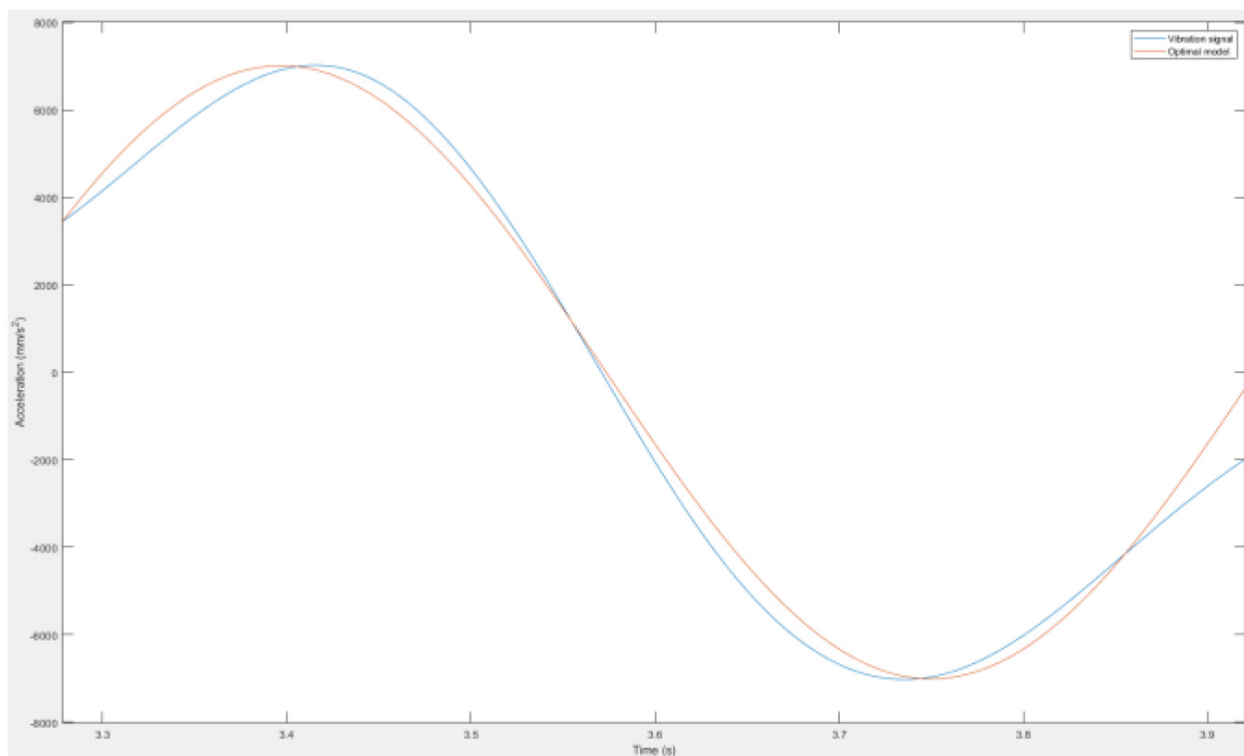


Figure 8. Vibration signal and optimal model.

Table 1. Waveform fitting results of rehabilitation training experiment.

Number	Start time	End time	Amplitude (A) (mm/s ²)	Velocity (W)	Correlation coefficient (P)
1	2.75	3.581	7070	6.7	0.9971
2	3.278	3.921	7010	8.9	0.9965
3	3.236	3.994	7140	7.7	0.9959
4	3.472	4.232	8180	7.9	0.9841
5	3.657	4.286	7900	9.4	0.9938
6	3.496	4.133	7500	8.7	0.9938
7	3.395	4.103	8160	8.5	0.994
8	2.722	3.441	6510	8.1	0.991
9	3.833	4.48	8050	8.8	0.9883
10	3.484	4.248	6560	6.8	0.9804
				AVG(P):	0.99149

correlation coefficients in 10 groups is from 0.9804 to 0.9971. And P is 0.99149 on average.

$AVG(P) = 0.99149 > 0.95$, shows very strong correlation between sinusoidal model and vibration signal. As a result, we can get a general expression of vibration signal in rehabilitation training:

$$f(x) = A \sin(\omega t + \varphi) \text{ mm/s}^2$$

A simple compound action generates a sinusoidal acceleration image. The complete movement of rehabilitation training can be divided into two stages. The schematic diagram of the physical process of rehabilitation training is shown in Figure 9. In stage I, in order to move, the contraction of muscles causes the palm to accelerate in the direction of movement and then reach the peak of acceleration in a short time. In this process, the speed of palm movement gradually reaches the maximum value. Then, in stage II, the muscles begin to generate acceleration in the opposite direction of motion to keep the palm in the designated position. This process is the same as stage I, but it is the opposite. Finally, the acceleration is reset to the static value, and the palm stops moving.

Wheelchair. The wheelchair experiment [Zhu, 2022f] uses a crawler robot to simulate the motion of wheelchairs because they have similar mechanical structures and movement modes. The robot used in this experiment is the mBot Ranger by makeblock. Similar to a wheel of a wheelchair, its tracks provide support and forward acceleration for the central part simultaneously. The vibration sensor is set on the robot's top tightly to collect vibration data when moving

forward, as shown in Figure 10(a). The direction of the vibration sensor's X-axis is the same as the forward direction of the robot. The experiment was carried out on the indoor smooth marble floor to decrease distractions by environmental factors like stone, debris, or uneven ground. The experiment environment is shown in Figure 10(b) and video S2 [Zhu, 2022d]. The distance between starting line and line is 2.30m, and there are no obstacles. The robot first stopped at the starting line. The robot controlled by a wireless app moved straight forward at 47cm/s. Then it stopped at the ending line. The collected data was downloaded and transported by the WiserVibe app. After that, all data was processed in Matlab software [Zhu, 2022g] (GNU octave is an open source alternative that may be able to perform similar functions).

The primary method of data analysis in wheelchair experiments is similar to the analysis in rehabilitation training experiments. The general expression of the model in the wheelchair experiment is needed to be found. First, a basic model through observation and experience is determined. Through waveform fitting, the value of the correlation coefficient is calculated so that the relationship between the wheelchair experiment signal and the assumed model can be judged and verified. The expression of the correlation coefficient is the same as the expression in the rehabilitation training experiment.

Before waveform fitting, we need to assume a basic model for vibration signal in the wheelchair experiment. The wave of the wheelchair experiment has an obvious beating vibration phenomenon. Beat vibration is caused by the superposition of the vibration generated by two forces with similar

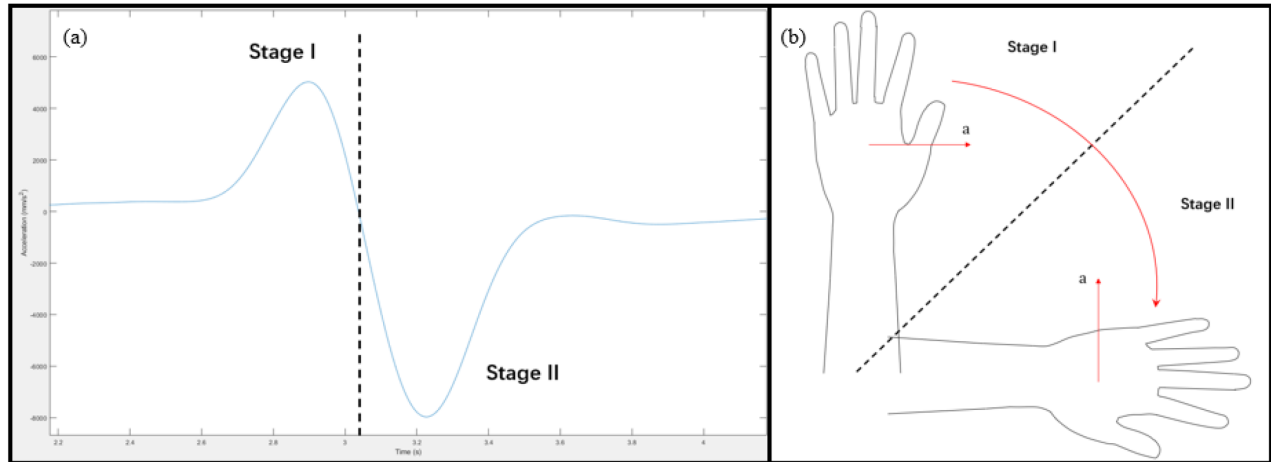


Figure 9. Stages in rehabilitation training (a) stages of waveform (b) schematic diagram of physical process.

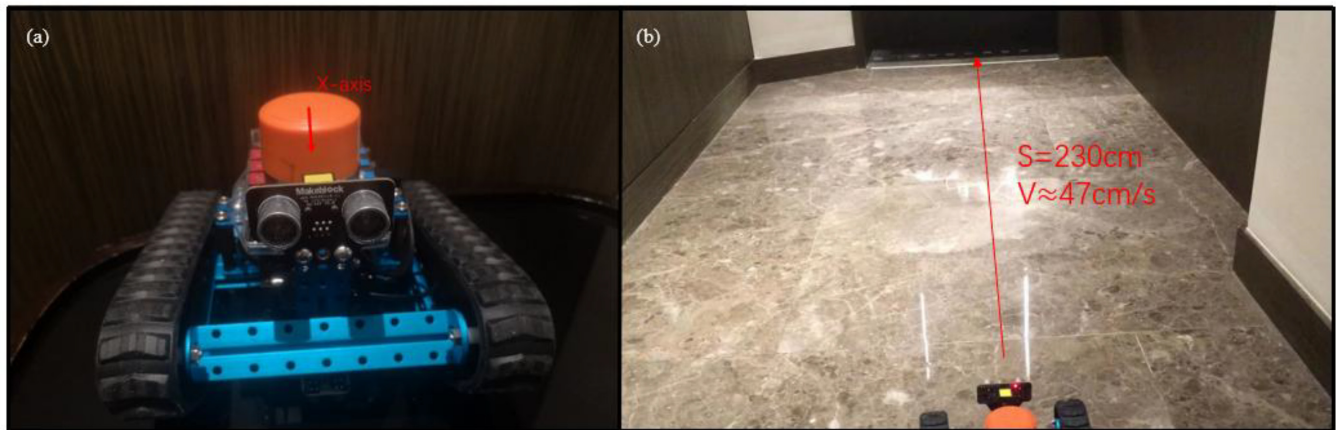


Figure 10. The wheelchair experiment (a) setup of vibration sensor on robot (b) simulation environment.

frequencies (excitation sources). Therefore, we assume that the vibration signal is the amplitude-modulated harmonic of two sinusoidal signals.

Figure 11 plots the vibration signal and related optimal model, which is the parameters group number 7 in Table 2. The optimal model is the parameters group with biggest correlation coefficient value. The model can match most of the parts in the vibration signal, including the fundamental frequency and the maximum and minimum range. However, the optimal result is still not perfect enough. As shown in the figure, the waveform of actual vibration data is not equispaced, but the optimal model is equispaced, leading to a specific deviation in the two waveforms. In addition, the maximum and minimum values of the waveform are usually not fixed but have certain randomness. These conditions will worsen the

fitting effect, resulting in the deviation between the model and the vibration signal. These phenomena can explain that the physical effect of the sensor is unstable. Therefore, the interference of vibration signal is obvious, and the difficulty of fitting will significantly increase.

The waveform fitting results of the wheelchair experiment for ten recording groups in total are shown in Table 2. The start time and end time are recorded. For each group, 3 seconds time domain segment is selected. The amplitude and angular speed for two signal components and corresponding correlation coefficient values are recorded. A1 and W1 are the amplitude and angular speed of first signal component. A2 and W2 are the amplitude and angular speed of second signal component. P is the correlation coefficient between physical signal and digital signal.

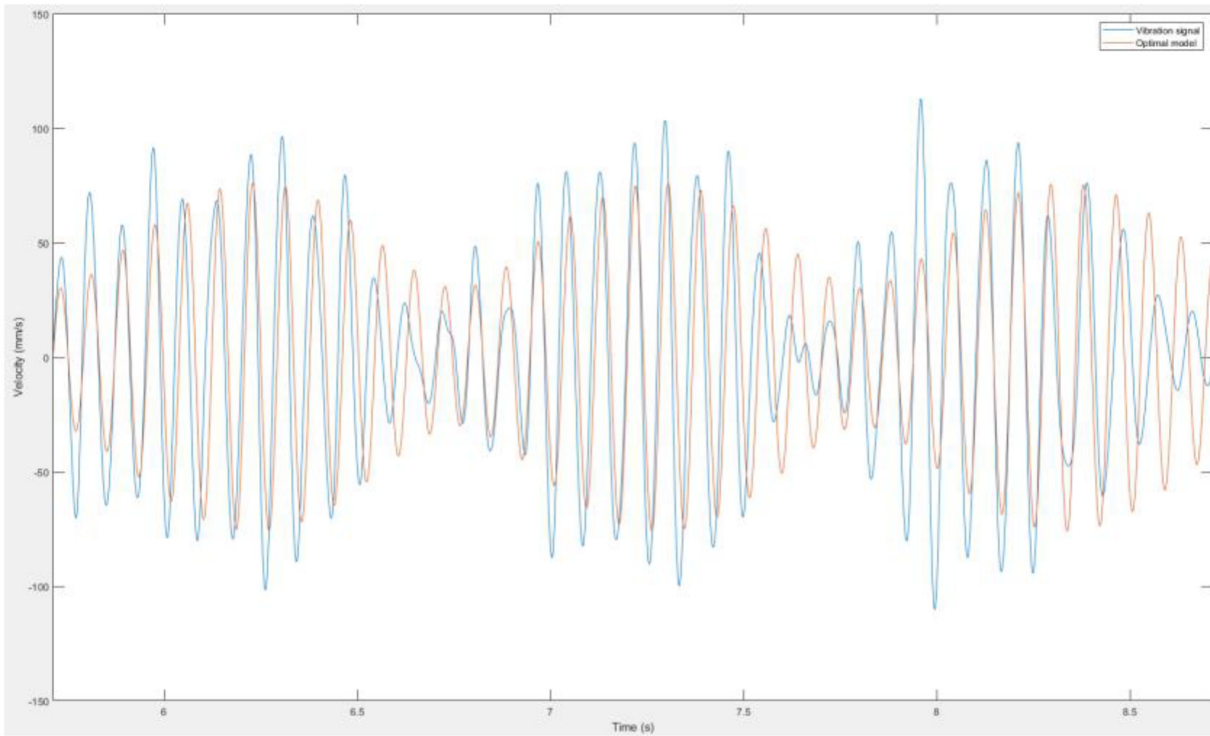


Figure 11. Comparison between vibration signal and optimal model.

Table 2. Waveform fitting results of wheelchair experiment.
(W – velocity, A – amplitude).

Number	Start time	End time	W1	W2	A1	A2	P
1	1.03	4.03	81	77	62	21	0.7309
2	4.21	7.21	82	76	50	40	0.744
3	2.09	5.09	80	78	51	34	0.7644
4	3.01	6.01	80	72	50	40	0.6979
5	3.595	6.595	79	76	70	27	0.6937
6	3.284	6.284	77	84	69	23	0.7427
7	5.713	8.713	76	70	53	23	0.7111
8	3.434	6.434	75	74	65	24	0.7993
9	2.07	5.07	73	76	54	50	0.7718
10	3.1	6.1	73	79	70	20	0.8415
						AVG(P)	0.74973

The average value of the correlation coefficient is 0.75973 > 0.7, which shows a moderately strong correlation. Therefore, we can explain the expression of the general model in the wheelchair experiment as follows:

$$f(x) = A_1 \sin(\omega_1 t + \varphi_1) + A_2 \sin(\omega_2 t + \varphi_2)$$

When compared to the correlation coefficient value in the rehabilitation training experiment (0.99), this result is not so convincing. After investigation, we found that there are three factors that affect the quality of the signal that decrease the accuracy of waveform fitting results: firstly, the actual physical factors are challenging to maintain stability on the

ground; secondly, it is challenging to keep the velocity of the equipment constant; thirdly, the coordination between wheels and core components is complex. Nevertheless, because of that, the test environment is more representative. In the natural environment, all kinds of interference and vibration of parts are usual. For example, the road surface that the wheelchair passes cannot be smooth; the speed of the wheelchair cannot be kept constant; at the same time, the wheelchair cannot be completely stable when moving forward since natural friction or collision of components in the wheelchair will also produce continuous vibration. The result of waveform fitting is more suitable for natural scenes.

It is known that the waveform collected by the vibration sensor will be sinusoidal in uniform motion. Then it can be inferred that the movement speed of the robot or wheelchair can be decomposed into two speeds in the same direction and of different amplitudes. In the experiment, the forward speed is approximately constant. So the robot's speed is decomposed, as shown on the left of Figure 12. During robot's movement, there will be vibration between the wheel and the main structure. This vibration acts on the main structure in the vertical direction, so it produces velocity V_{up} or V_{down} on the sensor. Therefore, for the sensor, the speed value is not constant after synthesis, and the speed waveform is not a stable single sinusoidal waveform. As a result, the speed detected by the sensor is recombined into $V_{forward}$ and $V_{additional}$. $V_{forward}$ is the approximate constant speed of robot's movement. $V_{additional}$ is the additional speed caused

by structural vibration. The combination of the two velocity generates the waveform in wheelchairs experiment.

Human fall. For the human fall experiment [Zhu, 2022a], the scenario is closed to a simple falling movement, which is affected by the human body and environment at the same time. The impact of falling depends on multiple factors like posture, speed, and height in actual cases. However, in experiments, we only need to set the simplest falling scenario that falls back vertically. In the experiment, the subject's height was 182cm, and weight was 73kg. The mattress used for the test is 50cm above the ground, so the actual drop is 132cm. At the beginning of this experiment, the subject wore the bracelet on his hand and placed the hand on the chest, standing and back towards the mattress. Then, the subject fell straight back on the mattress. The subject's back first came into contact with the mattress, causing an impact. At last, the subject keeps resting until the record is over. The complete experimental record is shown in Figure 13 and the video S3 [Zhu, 2022e].

It is essential to clear the parameters used in threshold detection in human fall experiments. As we know, the direction of human fall is random. It can be forward, backward or lateral. In the falling process, the direction changes in nearly all process periods. Therefore, the parameters used in data analysis must combine vibration signals of the three-axis in the vibration sensor. Refer to the thresholds commonly used in the experiment, in human fall experiment data analysis.

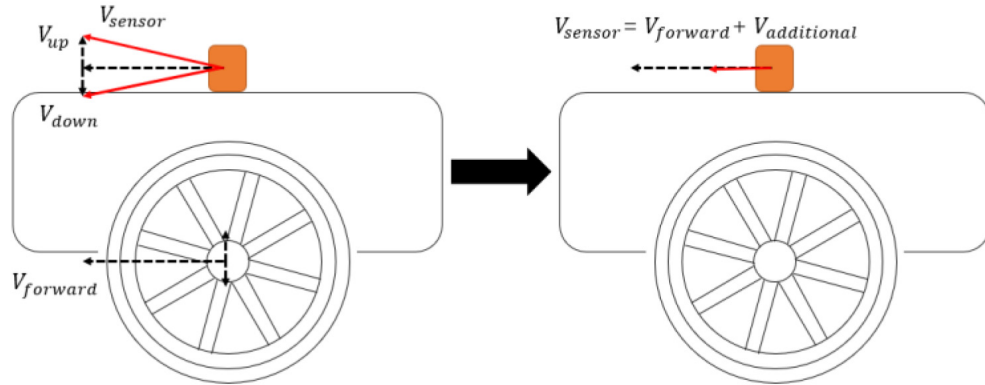


Figure 12. Schematic diagram of decomposition and synthesis of velocity.

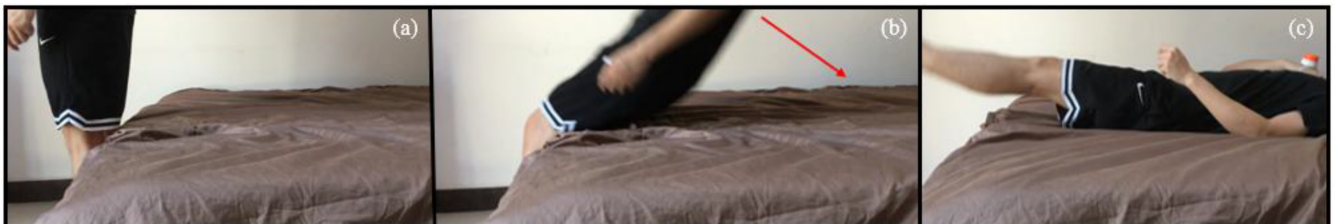


Figure 13. Complete falling process in experiment (a) standing (b) falling (c) impact.

The below parameters are used as thresholds to detect falls (Pan *et al.*, 2020):

SV is the total sum vector. It contains vibration signals from the X, Y, and Z axes in the vibration sensor, which correspond to acceleration values in the X, Y, and Z axis. SV is equal to the sum of the squares of the three axes under the root sign and is expressed below expression:

$$SV = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

SV_D is the dynamic sum vector. It also contains vibration signals from X, Y and Z axis. Different from SV, the acceleration signals $A_x^{(1)}$, $A_y^{(1)}$, $A_z^{(1)}$ are filtered. The vibration signal passed through a high pass filter with a cut-off frequency of 3Hz. The low-frequency part of the signal is removed, and the high-frequency part is retained. After that, SV_D follows the same mathematical expression as SV, and replaces A_x , A_y , A_z by $A_x^{(1)}$, $A_y^{(1)}$, $A_z^{(1)}$:

$$SV_D = \sqrt{(A_x^{(1)})^2 + (A_y^{(1)})^2 + (A_z^{(1)})^2}$$

The vertical acceleration dynamic quantity (BVA) is found in expression as shown in below:

$$BVA = \frac{SV^2 - SVD^2 - G^2}{G^2}$$

In which G is the gravitational acceleration constant. In this research, $G = 1\text{ g} = 10\text{ m/s}^2$.

In order to represent differences more intuitively, for one vibration data group in the human fall experiment, Figure 14 plots the waveform curve results of SV, SV_D , and BVA separately. For SV waveform curve, it contains signals from the three axes. However, the baseline is not zero and has a specific offset. In the curve of SV_D , the offset of the baseline is eliminated. Nevertheless, after the impact, the attenuation speed of signal amplitude is relatively slow. It may be misjudged as multiple impacts in the system. The curve of BVA is ideal. The baseline is 0 and steady. The amplitude changes obviously and violently when the impact occurs. When the impact disappears, the amplitude decays quickly. As a result, BVA is more suitable for automatically detecting human falls.

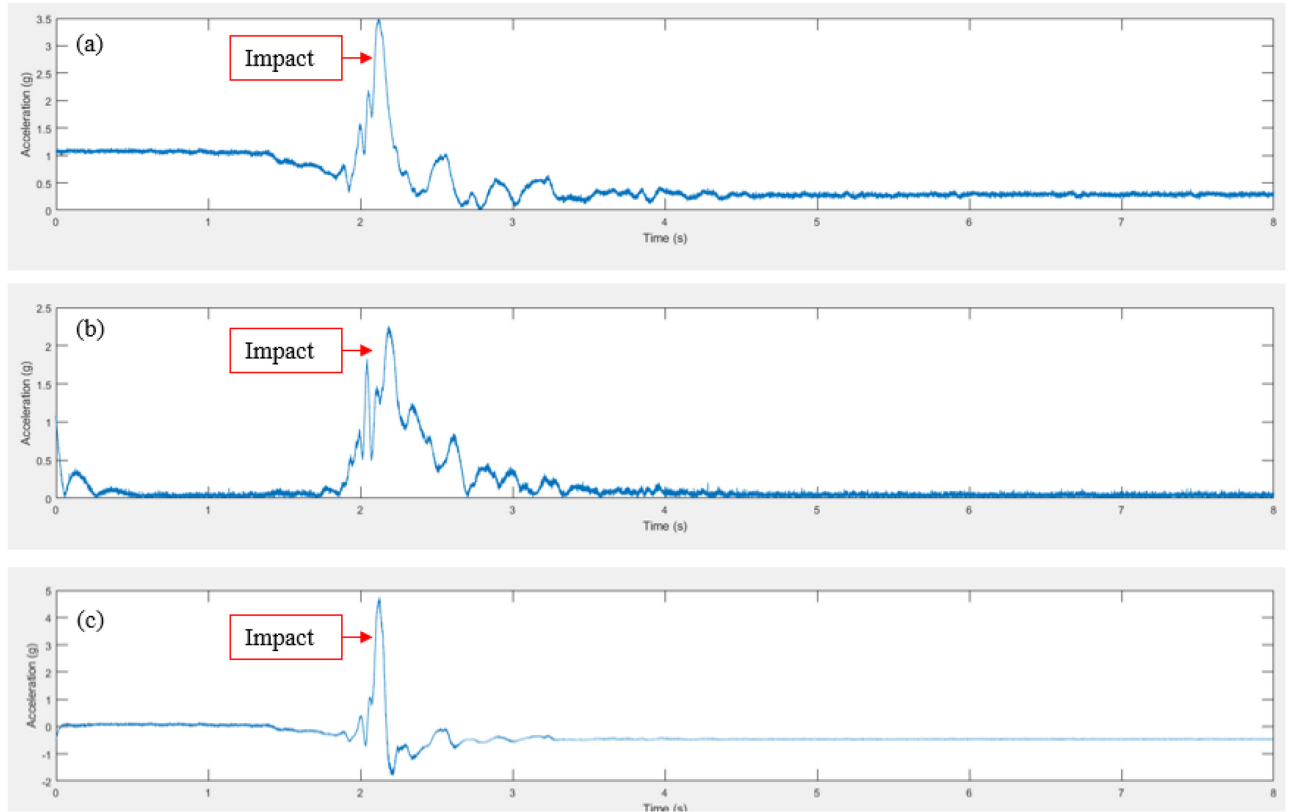


Figure 14. Human fall experiment curve (a) Sum vector (SV) curve and impact location (b) SV_D curve and impact location (c) BVA curve and impact location.

We counted the maximum values of three parameters in ten data groups to set the appropriate threshold. The maximum values are presented in the form of a scatter diagram, which represents the distribution of the maximum values of different parameters, as shown in Table 3 and Figure 15. Among SV , SV_D , and BVA , SV_D shows more stability in the maximum value. The maximum value of c shows great discreteness. When the subjects and falling postures are similar, the amplitude of BVA may have a maximum value difference of about four times. The discrete type of a is between the other two parameters. In short, although the BVA waveform curve

Table 3. Summary of maximum value in Human fall experiment (a) the maximum values of SV/g , (b) the maximum values of SV_D/g , (c) the maximum values of BVA/g .

No.	max (SV)/g	max (SV_D)/g	max (BVA)/g
1	4.3507	2.7856	6.7877
2	3.4879	2.2589	4.7468
3	3.5219	2.4222	3.887
4	5.6655	3.1547	11.4556
5	3.2179	2.5788	3.6548
6	5.5181	3.4011	11.8361
7	6.3379	3.6661	16.4974
8	4.2308	2.6375	6.5378
9	4.1741	2.3779	6.4415
10	5.0233	3.1527	10.4856

is in an ideal pattern, its threshold is hard to be determined. Using SV or SV_D to determine the threshold for detection can be easier to detect a human fall.

BVA curve is selected to represent better periodicity for analysis. After the experiment, the process of human fall can be divided into five stages. As shown in Figure 16, In the time domain waveform, the waveform has five stages from the beginning to the end of the fall. The stages from beginning to the ending are: standing, falling, impact, attenuation, and resting.

We can conduct mechanical and state analysis on the complete fall process by combining waveform results and a specific schematic diagram (Figure 17). In stage I, the subject stands and there is a balance between supporting force F_N and gravity G . In stage II, which is the falling process, the subject loses balance, supporting force F_N is less than gravity G so that generates downward acceleration. In the waveform, the acceleration begins to rise. In stage III, a huge crest appears in the waveform because the huge impact force F is generated, and it is much larger than G . Assuming that the speed decreases to 0 in time t after a person falls, according to the momentum theorem, the average impact force F in time t is:

$$(F - G)t = mv$$

$$F = \frac{mv}{t} + G$$

In which m is the weight of the subject, v is the speed of falling to ground, t is the time to reduce the speed to 0 after contacts with the ground. In most of cases, t is within 0.5s or even smaller because the elasticity of the human body is not great. Thus, impact force F is much greater than gravity G and cause a huge change in acceleration.

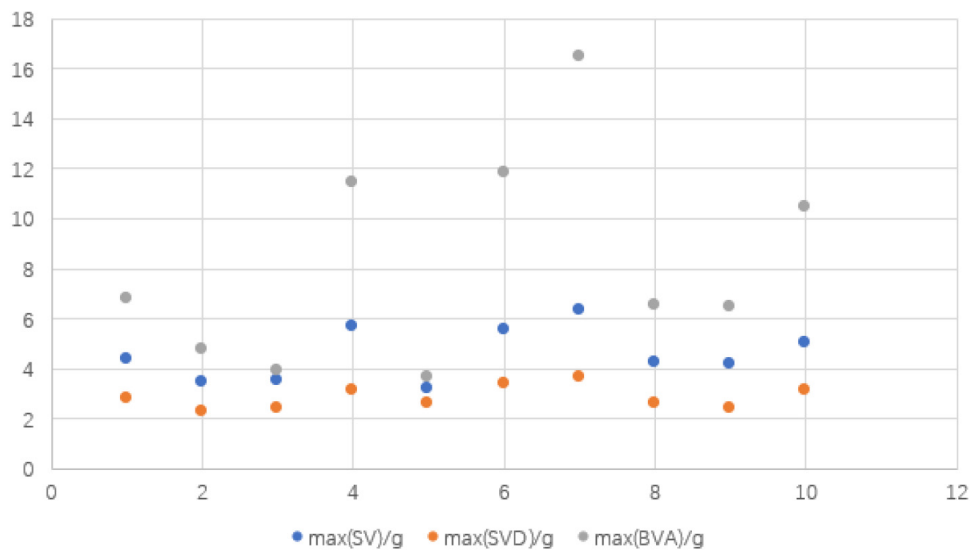


Figure 15. Distribution of (a) the maximum values of SV/g , (b) the maximum values of SV_D/g , (c) the maximum values of BVA/g .

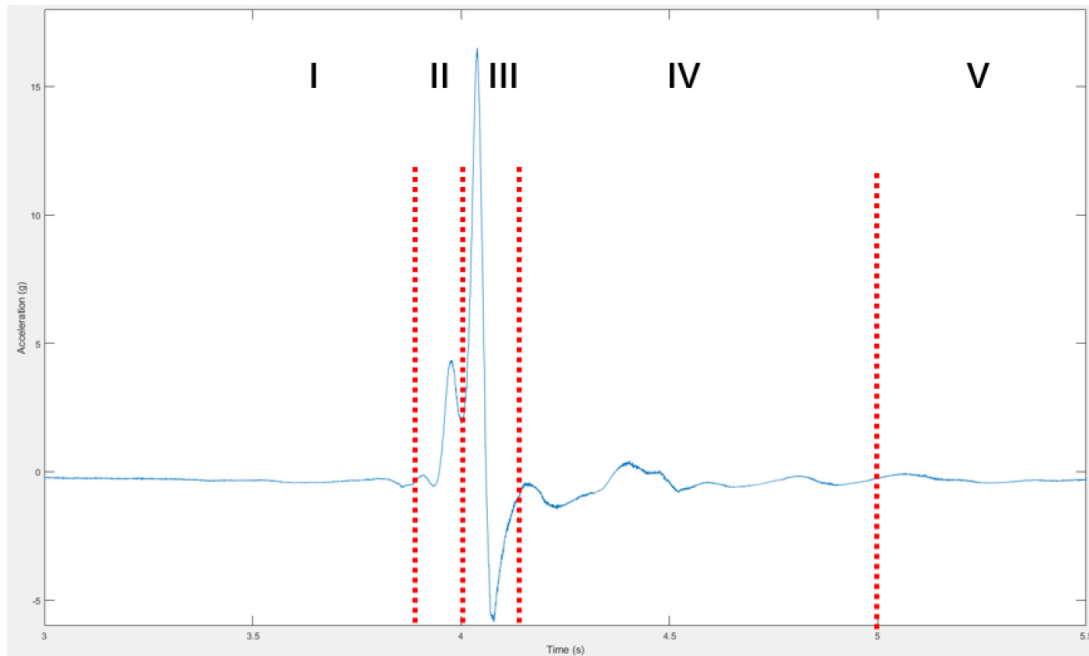


Figure 16. Stages in waveform of BVA (I) Standing (II) Falling (III) Impact (IV) Attenuation (V) Resting.

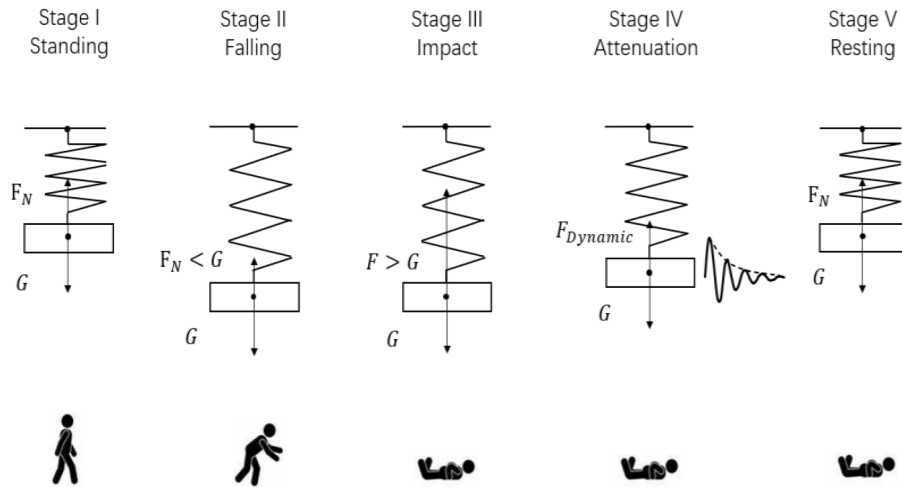


Figure 17. Physical analysis in human fall event.

However, at the same time, the human body is a collection of muscle tissue, joints, and bones, which still has a certain elasticity. The human body will not fully absorb the momentum generated in stage III. Instead, in stage IV, the human body absorbs part of the momentum while the other part is transformed into gradually decaying kinetic energy. This process is similar to the damped motion of a spring in its natural state. In the time domain waveform, the gradually decaying acceleration waveform is generated. After stage IV, the human body returns to equilibrium in stage V, and

F_N is equal to G . In some studies (Pascoletti *et al.*, 2019), the human body is modeled into multiple units, so that there is a more detailed biomechanical analysis of falls, and results is also more accurate.

Discussions

Although various of wearable devices and DT in healthcare have already been studied, the novelty of this study is a combination between DT and wearable healthcare devices, which offers personalization and data continuity in a low-cost

and efficient way. The originality of this research is beyond personalization in a particular device, because this research proposes a general DT-enabled system that suits mass healthcare devices development. This study not only evaluates sensor performance but also constructs a dynamic personalized paradigm through DTES. For example, in rehabilitation training scenarios, the DT model can adjust the mechanical parameters (such as vibration frequency) of virtual devices in real-time based on user motion data and synchronize them with physical devices. This bidirectional optimization capability upgrades personalization from static adaptation to dynamic response, significantly different from the fixed design logic of traditional wearable devices. Although DTES showed a high correlation ($\rho=0.991$) in rehabilitation training, the wheelchair experiment's $\rho=0.749$ indicates that there are challenges in model fitting in complex motion scenarios. This suggests that DTES needs to introduce environment-adaptive algorithms in the future.

For wearable devices, it is challenging to balance the wearing experience and the quality of signal acquisition. Shape matching is essential for physiological signal monitoring, especially long-term physiological monitoring. Using 3D printing to match wearable medical devices with human body structure has been a popular solution (Rachim & Park, 2021). A study (Abdollahi, 2018) introduced the concept of customizing wearable devices with 3D printing and carried out 3D printing for wearable devices for patients of different sizes. This study extends the customization of wearable medical devices to mass personalization to further promote the matching between wearable medical devices and patients. At present, flexible electrodes with mechanical flexibility are continuously developed (Lim *et al.*, 2020). A combination between this study and flexible electrodes will provide greater application ability in the future.

With this research, we tried to answer further research calls in mHealth (Hoque & Sorwar, 2017) and wearable patient monitoring (Baig *et al.*, 2017) for the elderly and disabled. As a fairly transdisciplinary exertion to noteworthy studies (Indrakumari *et al.*, 2021), this research lets us move forward in the related field. As wearable devices provide more personalization, better healthcare processes can be provided not only for patients, but also for their families, doctors, and all stakeholders. Therefore, this study is a significant construct for the success of personalized healthcare.

Conclusions

This research proposes a DT-enabled smart device system and develops a personalized wearable device for healthcare based on DT. It is a challenge for wearable device to suit different disease types of and physiological structures of patients. This study's novelty is to extend the customization method to meet specific medical needs to a general personalized application framework.

The long-term care of the elderly and the disabled has brought great pressure to the family and the government. Wearable medical system can bring convenience and relieve pressure

through remote monitoring. Internet of things enabled wearable devices for health can seamlessly collect users' physiological data at the front end and establish transmission channels between physiological data and the Internet. The DT can increase efficiency and reduce personalized production and manufacturing costs by combining entities and digital objects. The combination of the two provides considerable application value.

A DT-enabled smart system is proposed. DTES is designed to serve the personalization and data continuity for wearable device. It can connect various producers in design and manufacturing and produce personalized devices in a low-cost and efficient way. DTES aims to provide users with a high-quality signal acquisition, transmission, and data analysis, so it is very suitable for medical service scenarios.

This research designs a case study with three different scenarios, later designed and manufactured the triaxial vibration collection bracelet, and finally clarified its internet of things pattern. The three scenarios are human fall, wheelchair, and rehabilitation training separately. They are common in society and relate to different movement modes. Using solidworks software, a personalized prototype of a triaxial vibration collection bracelet is designed based on the subject's hand shape and personalized parameters. Then, it is printed by 3D printer and assembled. An experiment was carried out for each scenario, and the results were analyzed. The triaxial vibration collection bracelet prototype collected accurate movement signals for related movements. The motion signal parameters of the user and movement pattern in different scenarios were identified. As a result, a balance between user experience and signal quality was acquired. This study is limited by sample diversity and short-term experimental period. In the future, it is necessary to expand to multiple geographical populations and evaluate the impact of long-term wearing on data bias.

This study not only validated the personalized potential of wearable devices but also achieved a paradigm shift from 'passive adaptation' to 'active prediction' through the DTES framework. The virtual real collaboration capability of DT enables medical devices to dynamically adapt to user needs and provide scalable solutions for large-scale personalization, which is difficult to achieve with existing technologies. Future researches might focus on improving the application ability of real scenarios. The software application of DTES will be developed. At the same time, we will focus on the security of medical information in the process of signal transmission and information sharing and develop unique data protection methods to protect users' highly personal medical privacy in practical applications.

Data availability

Underlying data

Figshare: Human fall vibration data. <https://doi.org/10.6084/m9.figshare.20488905.v3> [Zhu, 2022a]

This project contains the following underlying data:

- Human fall-Group1.csv
- Human fall-Group2.csv
- Human fall-Group3.csv
- Human fall-Group4.csv
- Human fall-Group5.csv
- Human fall-Group6.csv
- Human fall-Group7.csv
- Human fall-Group8.csv
- Human fall-Group9.csv
- Human fall-Group10.csv

Figshare: Rehabilitation training vibration data. <https://doi.org/10.6084/m9.figshare.20488962.v2> [Zhu, 2022b]

This project contains the following underlying data:

- Rehabilitation training-Group1.csv
- Rehabilitation training-Group2.csv
- Rehabilitation training-Group3.csv
- Rehabilitation training-Group4.csv
- Rehabilitation training-Group5.csv
- Rehabilitation training-Group6.csv
- Rehabilitation training-Group7.csv
- Rehabilitation training-Group8.csv
- Rehabilitation training-Group9.csv
- Rehabilitation training-Group10.csv

Figshare: Wheelchair vibration data. <https://doi.org/10.6084/m9.figshare.20488980.v3> [Zhu, 2022c]

This project contains the following underlying data:

- Wheelchair-Group 1.csv
- Wheelchair-Group 2.csv
- Wheelchair-Group 3.csv

- Wheelchair-Group 4.csv
- Wheelchair-Group 5.csv
- Wheelchair-Group 6.csv
- Wheelchair-Group 7.csv
- Wheelchair-Group 8.csv
- Wheelchair-Group 9.csv
- Wheelchair-Group 10.csv

Extended data

Figshare: Video S1. <https://doi.org/10.6084/m9.figshare.20489034.v2> [Zhu, 2022d]

This project contains the following extended data:

- Experiment-rehabilitation training.mp4

Figshare: Video S2. <https://doi.org/10.6084/m9.figshare.20489043.v2> [Zhu, 2022e]

This project contains the following extended data:

- Experiment-wheelchair.mp4

Figshare: Video S3. <https://doi.org/10.6084/m9.figshare.20489016.v2> [Zhu, 2022f]

This project contains the following extended data:

- Experiment-human fall.mp4

Data are available under the terms of the [Creative Commons Zero “No rights reserved” data waiver](#) (CC0 1.0 Public domain dedication).

Software availability

Source code available from: <https://github.com/ZhengxuZhu/A-digital-twindigital-twin-enabled-wearable-device-for-customized-healthcare.git>

Archived source code at time of publication: <https://doi.org/10.5281/zenodo.7277927> [Zhu, 2022g]

License: MIT

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Reviewer Report 02 April 2025

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Bala Murugan 

Vellore Institute of Technology, Vellore, India

This research article proposes a digital twin (DT) and wearable medical device combination to provide personalized healthcare efficiently. The authors developed a DT-enabled smart system (DTES) for personalized design, manufacturing, and data tracking of a healthcare device prototype.

A triaxial vibration collection bracelet was designed and manufactured for three healthcare monitoring scenarios: rehabilitation training, wheelchair use, and fall detection. The results demonstrated the bracelet's potential for application in these scenarios.

Detailed Review

1. No sufficient details of methods and analysis provided to allow replication:

- The authors should provide more comprehensive descriptions of the experimental setup, including specific parameters, equipment details, and data processing steps.
- Detailed information about the materials used, the 3D printing process, and the assembly of the device should be included.
- The statistical methods should be described in more detail, with clear justifications for the chosen techniques.
- This is critical for the scientific soundness of the article.

2. The conclusions drawn adequately/partly supported by the results:

- The authors should ensure that all conclusions are directly and clearly supported by the presented data.
- If there are limitations to the data or analysis, these should be acknowledged and discussed.

- Additional analysis or data may be needed to strengthen the conclusions.

Overall Assessment

The article presents an interesting and potentially valuable application of digital twin technology in personalized healthcare. However, the lack of sufficient detail in the methods and analysis, as pointed out by the reviewers, is a significant concern.

Key Points to Address for Scientific Soundness

- **Detailed Methods:** Provide a thorough description of the experimental design, materials, procedures, and data processing.
- **Statistical Clarity:** Clearly explain the statistical methods used and justify their appropriateness.
- **Robust Conclusions:** Ensure that all conclusions are firmly supported by the data and acknowledge any limitations.

By addressing these points, the authors can significantly improve the scientific rigor and credibility of their research.

Is the work clearly and accurately presented and does it cite the current literature?

Partly

Is the study design appropriate and is the work technically sound?

Partly

Are sufficient details of methods and analysis provided to allow replication by others?

No

If applicable, is the statistical analysis and its interpretation appropriate?

Not applicable

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Digital twin, Internet of Things

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Reviewer Report 12 March 2025

<https://doi.org/10.21956/digitaltwin.19411.r28867>

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**Jun Cai**

Concordia University, Montreal, Canada

The manuscript presents a novel integration of Digital Twin (DT) technology with wearable medical devices to enhance personalized healthcare by enabling real-time monitoring, data continuity, and adaptability. It introduces a DT-enabled smart system (DTES) that facilitates the rapid design and customization of wearable medical devices through digital mapping and 3D printing, improving efficiency and personalization in healthcare solutions. Additionally, the study validates the proposed system through case studies in rehabilitation training, wheelchair use, and human fall detection, demonstrating its potential application in various healthcare monitoring scenarios.

However, the proposed approach has limited innovation beyond prior research. While the study claims to enhance personalization in healthcare, the discussion does not sufficiently connect the DT framework to personalization and instead focuses heavily on evaluating a wearable sensor built by the authors themselves. Thus, the study lacks a clear demonstration of how the proposed system fundamentally advances personalized healthcare beyond existing wearable medical device frameworks. Additionally, many of the technical components, such as wearable sensors, cloud-based data transmission, and 3D-printed medical devices, have already been explored extensively in prior works. Without a stronger emphasis on how DT uniquely enables personalization, the contribution remains incremental rather than groundbreaking.

Besides, while the study provides experimental case studies, the methodology lacks sufficient statistical rigor. All analyses are based on a specific sensor and thus lacks sufficient generality. In addition, since the sample size is small and there is no clear validation against standard medical devices or established benchmarks, the correlation coefficients do not sufficiently validate the practical effectiveness of the proposed system.

Given all these, I recommend rejecting the manuscript in its current form.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Partly

Are sufficient details of methods and analysis provided to allow replication by others?

Partly

If applicable, is the statistical analysis and its interpretation appropriate?

Partly

Are all the source data underlying the results available to ensure full reproducibility?

Partly

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Digital twin

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Reviewer Report 15 February 2025

<https://doi.org/10.21956/digitaltwin.19411.r28824>

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Shuo Gao 

Beihang University, Beijing, Beijing, China

This article explores the innovation of integrating Digital Twin (DT) technology with wearable medical devices to enhance personalized healthcare through real-time monitoring and customized health support. It suggests that with the use of IoT, Big Data, and AI, DT can improve device design, manufacturing, and healthcare services, thus advancing personalized medicine. By digitally mapping device inputs and outputs, DT enables efficient, tailored care for patients, offering significant potential for personalized healthcare development. In addition, all previous comments have been well addressed. My only concern is that a new paper published in Nature reviews electrical engineering pictured a roadmap on human body digital twins. I suggest authors to cite this paper and discuss the proposed paper content is designed for which level of human body digital twins.

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Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Yes

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: human body modelling, sensors.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Version 1

Reviewer Report 15 November 2023

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Sihan Huang

School of Mechanical Engineering, Beijing Institute of Technology, Beijing, Beijing, China

In this paper, by concerning the patients with chronic diseases and the elderly, a digital twin-enabled smart system is proposed for personalization in the design, manufacturing, and data tracking of a healthcare device prototype. However, the main problem to be solved in this paper is not clear. Do you attempt to develop a new DT-enabled smart systems, or a new wearable device for personalization, or produce a device, etc? Here are some other issues for your reference:

1. The details of the proposed method should be presented in the abstract to some extend.
2. The motivation and gap of this paper are not clear. Is there any existing study on digital twin-based wearable device?
3. It would be better to organized literature review section around digital twin and wearable

device for healthcare. The current state cannot support your idea firmly.

4. The methodology section does not provide enough details to support your innovation. The description on realizing DT-enabled smart device system does not show the unique features. In other words, there are mature products that can collect user data now, like smart watch. Why is your system or device special and good?
5. The main work of this paper looks like fall into internet of things enabled device. Then, why you name this work as digital twin-enabled device? Again, internet of things enabled device is not new.
6. The figures in this paper are low information density, which can not tell enough details of your idea, for example figure 1, figure 2, etc.

Is the work clearly and accurately presented and does it cite the current literature?

Partly

Is the study design appropriate and is the work technically sound?

Partly

Are sufficient details of methods and analysis provided to allow replication by others?

No

If applicable, is the statistical analysis and its interpretation appropriate?

Not applicable

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Smart manufacturing, digital twin

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.
